

Zain-ul-Abidin || Metallurgical and Materials Engineer

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Career Objective:

Keen engineer with valuable academic background and hands-on experience in manufacturing and testing, aspiring to diversify my skillset by contributing in Inspection and quality firms to become an inspection expert.

Industrial Experience:

Assistant Manager at Treet Corp Pvt. Ltd. (Jun 24- Current)

- I was assigned as an Assistant Manager at Physical Vapor Deposition Section, Gained Hands-on training of thin film deposition and characterization.
- Managing and optimizing DC sputtering processes for high-quality thin film Deposition.
- Carried out Cross-sectional SEM analysis to measure the Cr-Sputtering layer thickness on 1-micron Blade edge.

MTO at Treet Group of Companies (Jul 23- Jun 24)

- 6-month hands on Training in various departments, gained multidisciplinary insight.
- Performed Micro-Vickers hardness testing on different Steels using a **diamond pyramidal indenter**.
- Applied microstructural analysis to assess **phase transformation** after heat treatments according to ASTM E112.

Internship at Qadcast Pvt. Ltd.

- Selected for a project based internship known as LFM (Leaders for Manufacturing) Program.
 - Conducted XRF quantitative and qualitative elemental analysis for alloy verification, ensuring accurate detection of trace elements and major constituents, to assure customer requirement and quality control.
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Education

University of Engineering and Technology, Lahore

- Bachelor of Science in Metallurgical and Materials Engineering
(CGPA: 3.637/4.00) || Oct 2019 – Jun 2023 ||
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Research and Industrial Projects

Standard Recipe for Foundry Manufactured Products — Internship Project

- Derived a cost effective and Time saving recipe to procure Raw Materials for Foundry Manufactured Products.
- Gained Extensive Experience of Elemental Analysis (XRF) to predict the suitable raw materials composition for specific product.

3-D Printed Natural Fiber Reinforced PLA composites -- Final Year Project

- Improved the mechanical Properties of Additive manufactured PLA composites Through Jute Fiber Reinforcement (Team Lead), aimed to replace existing bio-medical scaffolds and heavy structural components.

Corrosion Study of martensitic steel – Workplace Project

- Conducted a Corrosion study to compare the impact of different FUCHS Anti-Rust oils on corrosion resistance of Martensitic steel in atmosphere and Humidity Chamber as per ASTM D2247.
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Courses

Inspection and Quality assurance || A Grade || Lab: A- Grade ||

- Theoretical & Applied experience of Mechanical Testing, Non Destructive testing and radiography testing to detect internal flaws in welding and Castings following the ASTM standards.

Characterization of Engineering Materials || B- Grade||

- Theoretical and Experiential learning in interpreting the results of advanced characterization techniques such as: XRD, SEM, EDX, FTIR and DSC.

Corrosion Control and Engineering || A- Grade || Lab: A- Grade ||

- Theoretical Insight on Corrosion types and phenomenon along with practical experience at corrosion testing such as: Salt Spray (ASTM B117), cyclic Corrosion test (ASTM G85), Electrochemical testing (ASTM G5)
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Awards & Certificates

- Won a merit Based scholarship
 - Arena Simulation Certificate (Udemy)
 - Functional Advanced Materials Course (Coursera)
 - Participated in Meta Expo 2022 for poster design
Winner of scavenger Hunt in Metalympics 2022
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Skills

Microsoft Skills

ImageJ software for grain size analysis

Echem Analyst Software

- To measure corrosion potential, current density and Tafel slopes from polarization curves|
- To analyze the Electrochemical Impedance Spectroscopy (EIS) data to determine Impedance of materials.